MATH 6112. FALL 2026. ASSIGNMENT 2.

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Let $D_p = \begin{pmatrix} I_p & 0 \\ 0 & 0 \end{pmatrix}$ and we use e_i to denote the i-th standard basis vector of $\mathbb{C}^n$, with 1 in the i-th coordinate and 0 elsewhere.

Problem 1

(a) $\Longrightarrow$: for any eigenvalue λ of B with eigenvector v, we have that $B^2v = \lambda^2 v = Bv = \lambda v$. Since $v \neq 0$, $\lambda^2 - \lambda = 0 \implies \lambda = 0, 1$. Therefore, diagonalization of B implies it is similar to D_p for some $p \in \{0\} \cup [n]$. It is sufficient to show that B is diagonalizable, that is $\dim\ker B + \dim\ker(I-B) = n$. Suppose $\text{rank}(B) = n - k$, then $\dim\ker B = k$. Since $\dim\ker(I - B) + \dim\text{Im}(I - B) = n$, we need to show $\dim\text{Im}(I - B) = k$. Since for any $x \in \mathbb{C}^n$, $x = Bx + (I - B)x$ and hence $\text{span}(\text{Im}B, \text{Im}(I - B)) = \mathbb{C}^n$. For $y \in \text{Im}B \cap \text{Im}(I - B)$, let $y = Bt_1 = (I - B)t_2$, we have that

$$0 = (B - B^2)t_2 = By = B^2t_1 = Bt_1 = y.$$

Therefore, $\text{Im}B \cap \text{Im}(I - B) = \{0\}$ and since $\text{span}(\text{Im}B, \text{Im}(I - B)) = \mathbb{C}^n$, $\dim\text{Im}B + \dim\text{Im}(I - B) = n$. We know $\dim\text{Im}B = \text{rank}(B) = n - k$, thus, $\dim\text{Im}(I - B) = k$. In conclusion, B is diagonalizable.
$\Longleftarrow$: similarity implies that there exists $V \in \mathbb{C}^{n\times n}$ such that $B = V^{-1}D_pV$. Since $D_p^2 = D_p$, $B^2 = V^{-1}D_pVV^{-1}D_pV = V^{-1}D_pV = B$.
(b) Suppose $V = [v_1, \cdots, v_n]$. Since $V^{-1}BV = D_p$, we have that $Bv_i = v_i$ for $i \in [p]$ and $Bv_i = 0$ for $i > p$. For any vector $v \in \mathbb{C}^n$, suppose $v = \sum_{i=1}^n \lambda_i v_i$ since $\text{span}(V) = \mathbb{C}^n$ and we have that

$$Bv = \sum_{i=1}^n \lambda_i Bv_i = \sum_{i=1}^p \lambda_i Bv_i \in \text{span}\{v_1, \ldots, v_p\}.$$

Together with $Bv_i = v_i$ for $i \in [p]$, B projects onto the subspace $\mathcal{V} = \text{span}\{v_1, \ldots, v_p\}$.
(c) Note that the plane $2x_1 - 3x_2 + x_3 = 0$ is

$$\text{span}\left\{ \begin{pmatrix} 0 \\ 1 \\ 3 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix} \right\}.$$

Define

$$V = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 1 \\ 3 & 2 & 0 \end{pmatrix} \quad \text{and} \quad B = VD_2V^{-1} = \frac{1}{3}\begin{pmatrix} 3 & 0 & 0 \\ 2 & 0 & 1 \\ 0 & 0 & 3 \end{pmatrix}.$$

By Part **(b)**, B projects onto the subspace $2x_1 - 3x_2 + x_3 = 0$ and $B^* \neq B$. Take Gram-Schmidt Orthogonalization of V, we have that

$$U = \frac{1}{\sqrt{70}} \begin{pmatrix} 0 & -5\sqrt{2} & -2\sqrt{5} \\ \sqrt{7} & -3\sqrt{2} & 3\sqrt{5} \\ 3\sqrt{7} & \sqrt{2} & -\sqrt{5} \end{pmatrix} \quad \text{and} \quad \text{span}(U_1, U_2) = \{x \in \mathbb{R}^3 \mid 2x_1 - 3x_2 + x_3 = 0\},$$

where we use U_i to denote the i-th column of U. Take $B' = UD_2U^* = \frac{1}{14}\begin{pmatrix} 10 & 6 & -2 \\ 6 & 5 & 3 \\ -2 & 3 & 13 \end{pmatrix}$. By Part **(b)**, B' projects onto the subspace $2x_1 - 3x_2 + x_3 = 0$ and $(B')^* = B'$.

(d) Take $V = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$ and $B := VD_1V^{-1} = \begin{pmatrix} 1 & -t \\ 0 & 0 \end{pmatrix}$. By Part **(b)**, B projects onto $\text{span}\{e_1\}$ and $B^\top \neq B$. We have that $\|B\|_2^2 \geq \|B(e_1 - e_2)\|_2^2 / 2 = (t+1)^2/2$ and take $t \to \infty$, we get arbitrarily large 2-norm of B.

Problem 2

(a) Since A is normal, there exists a unitary transformation U such that $A = U\Lambda U^*$ where $\Lambda = \text{diag}\{\lambda_i\}_{i\in[n]}$ is the diagonal matrix of eigenvalues. Let $U = [u_1, \cdots, u_n]$. For $x \in \mathbb{C}^n$, suppose $x = \sum_{i=1}^n \mu_i u_i$ with $\sum_{i=1}^n |\mu_i|^2 = 1$ and we have that

$$x^*Ax = \sum_{i,j=1}^n \overline{\mu_i}\mu_j(u_i^*Au_j) = \sum_{i=1}^n |\mu_i|^2 \lambda_i \in \text{Co}(\sigma(A)).$$

Hence, $\mathcal{F}(A) \subseteq \text{Co}(\sigma(A))$. Fix $x = \sum_{i=1}^n \gamma_i\lambda_i \in \text{Co}(\sigma(\text{A}))$ with $\sum_{i=1}^n \gamma_i = 1$ and $\gamma_i \geq 0$ for $i \in [n]$, take $\mu_i = \sqrt{\gamma_i}$ and $y = \sum_{i=1}^n \mu_i u_i$. We have that $\|y\|_2 = 1$ and $y^*Ay = x$. Therefore, $\text{Co}(\sigma(A)) \subseteq \mathcal{F}(A)$. In conclusion, $\text{Co}(\sigma(A)) = \mathcal{F}(A)$.

(b) $\Longrightarrow$: A is Hermitian implies all the eigenvalues of A are real. By Part **(a)**, $\mathcal{F}(A)$ is a segment of the real line.

$\Longleftarrow$: for any $k \in [n]$, $e_k^\top Ae_k = A_{kk} \in \mathcal{F}(A) \subseteq \mathbb{R}$. Fix $k, j \in [n]$ with $k \neq j$. We have that

$$\frac{1}{2}(e_k + e_j)^\top A(e_k + e_j) \in \mathcal{F}(A) \subseteq \mathbb{R} \implies A_{kj} + A_{jk} \in \mathbb{R},$$

since $A_{kk}, A_{jj} \in \mathbb{R}$. Suppose $A_{kj} = \alpha_1 + \beta i$ and $A_{jk} = \alpha_2 - \beta i$ where $\alpha_1, \alpha_2, \beta \in \mathbb{R}$. We have that

$$\frac{1}{2}(e_k - ie_j)^\top A(e_k + ie_j) \in \mathcal{F}(A) \subseteq \mathbb{R} \implies i(A_{kj} - A_{jk}) \in \mathbb{R} \implies (\alpha_1 - \alpha_2)i \in \mathbb{R} \implies \alpha_1 = \alpha_2,$$

which implies $A_{kj} = \overline{A_{jk}}$. Therefore, A is Hermitian.

The endpoints of $\mathcal{F}(A)$ are $\lambda_{\min}(A)$ and $\lambda_{\max}(A)$.

(c) Define $P = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. For any $x \in \mathbb{C}^2$ with $\|x\|_2 = 1$, we can write $x = (\cos\alpha e^{i\theta_1}, \sin\alpha e^{i\theta_2})$, where $\alpha \in [0, \frac{\pi}{2}]$ and $\theta_1, \theta_2 \in [0, 2\pi)$, by polar transformation. We have that $x^*Px = \overline{x_1}x_2 = \frac{1}{2}\sin(2\alpha)e^{i(\theta_2 - \theta_1)}$. Taking over $\alpha \in [0, \frac{\pi}{2}]$ and $\theta_1, \theta_2 \in [0, 2\pi)$, we conclude that $\mathcal{F}(P)$ is a disk with radius $\frac{1}{2}$ and center 0.

It is clear that $\mathcal{F}(A) = \mathcal{F}(P + I) = 1 + \mathcal{F}(P)$ and $r(A) \leq 1 + \frac{1}{2}$ and the equality can be achieved since $\mathcal{F}(P)$ is a disk. In the same way, $\mathcal{F}(A^2) = \mathcal{F}(2P + I) = 1 + 2\mathcal{F}(P)$ and $r(A^2) = 1 + 2 \times \frac{1}{2}$.

Therefore, $r(A) = 3/2$ and $r(A^2) = 2$.

(d) Nonnegativity and positive homogeneity of numerical radius follow from the nonnegativity and positive homogeneity of the norm of complex number. If $r(A) = 0$, then $\mathcal{F}(A) = \{0\}$ and $A = 0$. The triangle inequality follows from the following inequality:

$$\begin{aligned} r(A+B) &= \max_{\|x\|_2=1} |x^*(A+B)x| \le \max_{\|x\|_2=1} |x^*Ax| + |x^*Bx| \\ &\le \max_{\|x\|_2=1} |x^*Ax| + \max_{\|y\|_2=1} |y^*Ay| = r(A) + r(B). \end{aligned}$$

Hence, $r(\cdot)$ is a norm. However, consider $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$. We have that

$$r(AB) \ge |e_1^\top ABe_1| = 1.$$

By AM-GM inequality,

$$\forall x \in \mathbb{C}^2 \text{ with } \|x\|_2 = 1, |\overline{x_1}x_2| = |x_1|\,|x_2| \le \frac{1}{2}\left(|x_1|^2 + |x_2|^2\right) = \frac{1}{2} \implies r(A) \le \frac{1}{2}.$$

In the same way, $r(B) \le \frac{1}{2}$. Therefore, $r(AB) = 1 > \frac{1}{4} \ge r(A)r(B)$ and r is not a consistent matrix norm.

Problem 3

$$\begin{aligned} \begin{pmatrix} A & 0 \\ I & B \end{pmatrix} &\xrightarrow{\text{swap columns}} \begin{pmatrix} A & 0 \\ I & B \end{pmatrix}\begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix} = \begin{pmatrix} 0 & A \\ B & I \end{pmatrix} \\ &\xrightarrow{\text{Add } (-A\times\text{row 2}) \text{ to row 1}} \begin{pmatrix} I & -A \\ 0 & I \end{pmatrix}\begin{pmatrix} 0 & A \\ B & I \end{pmatrix} = \begin{pmatrix} -AB & 0 \\ B & I \end{pmatrix} \\ &\xrightarrow{\text{Multiply row 1 by } -1} \begin{pmatrix} -I & 0 \\ 0 & I \end{pmatrix}\begin{pmatrix} -AB & 0 \\ B & I \end{pmatrix} = \begin{pmatrix} AB & 0 \\ B & I \end{pmatrix}. \end{aligned}$$

Therefore,

$$M = \begin{pmatrix} -I & 0 \\ 0 & I \end{pmatrix}\begin{pmatrix} I & -A \\ 0 & I \end{pmatrix} = \begin{pmatrix} -I & A \\ 0 & I \end{pmatrix} \quad \text{and} \quad W = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}.$$

By properties in the problem statement, we claim that

$$\begin{aligned} \det(A)\det(B) &= \det\begin{pmatrix} A & 0 \\ I & B \end{pmatrix} = (-1)^n \det\begin{pmatrix} 0 & A \\ B & I \end{pmatrix} = (-1)^n \det\begin{pmatrix} -AB & 0 \\ B & I \end{pmatrix} \\ &= (-1)^{2n} \det\begin{pmatrix} AB & 0 \\ B & I \end{pmatrix} = \det(AB). \end{aligned}$$

Problem 4

We first show that for any matrix $A \in \mathbb{F}^{n\times n}$, $\|A^*\|_2 = \|A\|_2$:

$$\|A\|_2 = \sup_{\|x\|_2=1} \|Ax\|_2 = \sup_{\|x\|_2=\|y\|_2=1} |\langle Ax, y\rangle| = \sup_{\|x\|_2=\|y\|_2=1} |\langle x, A^*y\rangle| = \sup_{\|y\|_2=1} \|A^*y\|_2 = \|A^*\|_2.$$

Therefore, we have that $\left\|(A^*)^{-1}\right\|_2 \|A-B\|_2 < 1$. Suppose B is not invertible, that is, there is a vector $v \in \mathbb{F}^n$ such that $\|v\|_2 = 1$ and $Bv = 0$. Hence, we have that

$$\begin{aligned}\left\|(A^*)^{-1}\right\|_2 \|A-B\|_2 &\geq \left\|(A^*)^{-1}v\right\|_2 * \|(A-B)v\|_2 = \left\|(A^*)^{-1}v\right\|_2 * \|Av\|_2 \\ &\geq \left|\left\langle (A^*)^{-1}v, Av\right\rangle\right| = \left|v^*(A^{-1}A)v\right| = 1,\end{aligned}$$

which contradicts $\left\|(A^*)^{-1}\right\|_2 \|A-B\|_2 < 1$.

Problem 5

Simple observation shows that

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} I & 0 \\ CA^{-1} & D - CA^{-1}B \end{pmatrix} \begin{pmatrix} I & -B \\ 0 & I \end{pmatrix}^{-1} \begin{pmatrix} A^{-1} & 0 \\ 0 & I \end{pmatrix}^{-1}.$$

Therefore, by Problem **(3)**, we have that

$$\begin{aligned}\det(M) &= \det \begin{pmatrix} I & 0 \\ CA^{-1} & D - CA^{-1}B \end{pmatrix} \det \begin{pmatrix} I & -B \\ 0 & I \end{pmatrix}^{-1} \det \begin{pmatrix} A^{-1} & 0 \\ 0 & I \end{pmatrix}^{-1} \\ &= \det(D - CA^{-1}B) \det(A^{-1})^{-1} \\ &= \det(A) \det(D - CA^{-1}B) = \det(AD - ACA^{-1}B) && (\textbf{Problem } (3)) \\ &= \det(AD - CAA^{-1}B) = \det(AD - CB), && (AC = CA)\end{aligned}$$

where the second equality follows from $\det(A^{-1}) = \det(A)^{-1}$ and the determinant of a block matrix.

For the counterexample, let us consider $n = 2$ and

$$M = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 1 & 2 & 0 & 1 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 2 \end{pmatrix}.$$

We have that $\det(M) = 1$ while $\det(AD - CB) = 0$.

Suppose A is not invertible. The eigenvalues of A are $\lambda_1, \cdots, \lambda_n$ and let

$$m = \begin{cases} 1 & , \lambda_1 = \cdots = \lambda_n = 0 \\ \min_{i:\lambda_i \neq 0} |\lambda_i| & , \text{otherwise} \end{cases}.$$

Take $t \in (0, m)$, we have that $A + tI$ is invertible:

$$\exists V \in C^{n\times n} : A = V^{-1}\left(R + \operatorname{diag}\{\lambda_i\}_{i\in[n]}\right)V \implies A + tI = V^{-1}\left(R + \operatorname{diag}\{\lambda_i + t\}_{i\in[n]}\right)V,$$

where R is some strict upper triangular matrix and $\lambda_i + t \neq 0$ for $i \in [n]$. And also, $(A + tI)C = AC + tC = CA + tC = C(A + tI)$. Then by the previous part, we have that

$$\det \begin{pmatrix} A + tI & B \\ C & D \end{pmatrix} = \det(AD + tD - CB).$$

Note that $\det \begin{pmatrix} A + tI & B \\ C & D \end{pmatrix}$ is a polynomial function with respect to t, and thereby continuous. Taking t tends to 0, we conclude that

$$\det \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \lim_{t\to 0^+} \det \begin{pmatrix} A + tI & B \\ C & D \end{pmatrix} = \lim_{t\to 0^+} \det(AD + tD - CB) = \det(AD - CB).$$

Problem 6

(a) Diagonalization of A holds since A is Hermitian. Let U, V be unitary matrices satisfying that

$$U^*AU = V^*AV = \begin{pmatrix} \mu_1 I_{n_1} & & \\ & \ddots & \\ & & \mu_p I_{n_p} \end{pmatrix} \quad \text{and} \quad W = U^*V = \begin{pmatrix} W_{11} & \cdots & W_{1p} \\ \vdots & \ddots & \vdots \\ W_{p1} & \cdots & W_{pp} \end{pmatrix}.$$

Let $D = \mathrm{diag}\{\mu_i I_{n_i}\}_{i\in[p]}$. Following from the above equality, we have that $WD = DW$, that is,

$$\begin{pmatrix} \mu_1 W_{11} & \cdots & \mu_1 W_{1p} \\ \vdots & \ddots & \vdots \\ \mu_p W_{p1} & \cdots & \mu_p W_{pp} \end{pmatrix} = \begin{pmatrix} \mu_1 W_{11} & \cdots & \mu_p W_{1p} \\ \vdots & \ddots & \vdots \\ \mu_1 W_{p1} & \cdots & \mu_p W_{pp} \end{pmatrix}.$$

For $i \neq j$, $\mu_i \neq \mu_j$ but $\mu_i W_{ij} = \mu_j W_{ij}$, implying $W_{ij} = 0$. Therefore $W = \mathrm{diag}\{W_{ii}\}_{i\in[p]}$. Since $\mathrm{diag}\{W_{ii}^* W_{ii}\}_{i\in[p]} = W^*W = V^*UU^*V = I$, we have that $W_{ii}^* W_{ii} = I_{n_i}$ for any $i \in [p]$. This completes the proof.

(b) The proof is exactly the same. Diagonalization of A holds since A is symmetric. Let U, V be orthogonal transformations and $U^\top AU = V^\top AV = \mathrm{diag}\{\mu_i I_{n_i}\}_{i\in[p]}$. Let $D = \mathrm{diag}\{\mu_i I_{n_i}\}_{i\in[p]}$ and $W = U^\top V$. We have that $DW = WD$, implying $W = \mathrm{diag}\{W_{ii}\}_{i\in[p]}$ is diagonal block matrix. Since $W = U^\top V$, W is orthogonal and thus, W_{ii} is orthogonal for $i \in [p]$. This completes the proof.